

Computational Imaging

BMED365: Topic List Y01–Y11

BMED365

Department of Biomedicine, University of Bergen

Spring 2026

Overview

- 1 Fundamentals of Computational Imaging
- 2 Imaging Mass Cytometry
- 3 AI in Medical Image Analysis
- 4 Advanced Topics

Y01: Explain fundamental concepts in computational imaging

What is Computational Imaging?

Definition

Computational imaging combines **optics**, **sensors**, and **algorithms** to create images that go beyond traditional photography – extracting information invisible to conventional cameras.

Key Characteristics:

- Joint design of hardware and software
- Computational reconstruction
- Enhanced resolution, depth, spectrum
- Multi-dimensional data capture

Applications in Biomedicine:

- MRI, CT, PET reconstruction
- Microscopy (confocal, light-sheet)
- Optical coherence tomography
- Imaging mass cytometry

The Future

AI and deep learning are revolutionizing computational imaging – enabling faster acquisition, higher resolution, and automated analysis.

Y02: Describe the imaging pipeline

The Computational Imaging Pipeline



1. Acquisition:

- Physical signal capture
- Sensor characteristics (noise, dynamic range)
- Sampling and discretization

2. Processing & Analysis:

- Noise reduction, artifact correction
- Feature extraction
- AI-based interpretation

Y03: Understand digital image representation

How are images represented digitally?

2D Images (Pixels):

- Matrix of intensity values
- Grayscale: single channel (0–255)
- Color: RGB (3 channels)
- Resolution: pixels $\times$ pixels

3D Volumes (Voxels):

- Stack of 2D slices
- Common in MRI, CT
- Isotropic vs. anisotropic
- Memory: N^3 voxels

Key Concepts:

- **Spatial resolution:** Detail level
- **Intensity resolution:** Bit depth
- **Temporal resolution:** Frame rate
- **Spectral resolution:** Wavelengths

Medical Formats

DICOM: Standard for medical images

NIfTI: Neuroimaging format

TIFF: Microscopy standard

Y04: Explain basic image processing operations

Fundamental Image Processing Operations

Point Operations:

- Brightness/contrast
- Histogram equalization
- Thresholding
- Intensity normalization

Spatial Filtering:

- Smoothing (Gaussian)
- Sharpening (Laplacian)
- Edge detection (Sobel)
- Median filter (noise)

Morphological:

- Erosion/dilation
- Opening/closing
- Hole filling
- Skeleton extraction

Python Libraries

scikit-image: General image processing

OpenCV: Computer vision

SimpleITK: Medical image processing

scipy.ndimage: N-dimensional images

Y05: Describe Imaging Mass Cytometry (IMC) and its applications

What is Imaging Mass Cytometry?

Definition

IMC combines **immunohistochemistry** with **mass spectrometry** to simultaneously visualize 40+ protein markers in tissue sections at subcellular resolution.

How it works:

- 1 Antibodies tagged with metal isotopes
- 2 Laser ablates tissue pixel-by-pixel
- 3 Mass spectrometer detects metals
- 4 Reconstruct multiplexed images

Applications:

- Tumor microenvironment
- Immune cell profiling
- Spatial proteomics
- Drug response studies
- Biomarker discovery

Key Advantage

Unlike fluorescence microscopy (limited to 4–6 markers), IMC enables **highly multiplexed** analysis

Y06: Explain how IMC enables multiplexed tissue analysis

Multiplexed Analysis with IMC

Why multiplexing matters:

- Single-marker analysis misses complexity
- Cell phenotypes require multiple markers
- Spatial context is crucial
- Understand cell–cell interactions

Typical panel design:

- Structural markers (DNA, actin)
- Cell type markers (CD45, CD3, CD8)
- Functional markers (Ki67, PD-L1)
- Signaling markers (pSTAT3)

Analysis pipeline:

- 1 Image preprocessing
- 2 Cell segmentation
- 3 Feature extraction
- 4 Clustering / phenotyping
- 5 Spatial analysis

Tools

steinbock: IMC preprocessing

CellProfiler: Cell segmentation

Squidpy: Spatial analysis

Y07: Understand the role of AI in medical image analysis

AI is transforming medical imaging

Key AI Applications:

- **Detection:** Find lesions, tumors
- **Segmentation:** Delineate structures
- **Classification:** Diagnose conditions
- **Quantification:** Measure volumes
- **Registration:** Align images
- **Reconstruction:** Enhance quality

Why deep learning excels:

- Learns features automatically
- Handles high-dimensional data
- Generalizes across patients
- Processes images rapidly

Clinical Impact

FDA has approved 500+ AI medical devices (2023), majority in radiology.

Y08: Describe image segmentation approaches

Classical vs. Deep Learning Segmentation

Classical Methods:

- Thresholding (Otsu)
- Region growing
- Watershed algorithm
- Active contours (snakes)
- Atlas-based methods

Pros: Interpretable, no training

Cons: Limited accuracy, tuning needed

Deep Learning Methods:

- U-Net (encoder-decoder)
- nnU-Net (self-configuring)
- Attention U-Net
- TransUNet (transformers)
- SAM (foundation model)

Pros: State-of-the-art accuracy

Cons: Requires labeled data

Best Practice

Use **nnU-Net** as a strong baseline – it automatically adapts to your dataset and often achieves near-optimal performance.

Y09: Know about MedSAM

Segment Anything in Medical Images (MedSAM)

What is MedSAM?

A **foundation model** for medical image segmentation, trained on 1.5 million image-mask pairs across 10 imaging modalities.

Key Features:

- Based on Meta's SAM architecture
- Promptable segmentation
- Works across modalities
- Zero-shot generalization
- Interactive refinement

Modalities supported:

- CT, MRI, X-ray
- Ultrasound, PET
- Dermoscopy, endoscopy
- Fundus, OCT
- Microscopy

Practical Use

MedSAM enables rapid annotation and can be fine-tuned for specific tasks.

Y10–Y11: LLMs and Multimodal Imaging

Y10: LLMs in Medical Imaging

Emerging applications:

- Report generation from images
- Visual question answering
- Image-text retrieval
- Clinical decision support

Key models:

- GPT-4V for radiology
- Med-PaLM M (Google)
- LLaVA-Med
- BiomedCLIP

Y11: Multimodal Imaging

Combining modalities:

- MRI + PET (anatomy + function)
- CT + SPECT (structure + perfusion)
- Histology + IMC (morphology + proteins)

Challenges:

- Registration across modalities
- Different resolutions/scales
- Fusion strategies
- Interpretation complexity

Future Direction

Foundation models trained on multimodal medical data will enable unified analysis across imaging,

Summary: Computational Imaging

Key Learning Points:

- **Y01–Y04:** Fundamentals – imaging pipeline, digital representation, processing
- **Y05–Y06:** Imaging Mass Cytometry – multiplexed tissue analysis
- **Y07–Y09:** AI in imaging – segmentation, MedSAM
- **Y10–Y11:** Advanced – LLMs, multimodal imaging

Lab 4 Notebooks

00-test-installation	Verify environment
01-imaging-intro	Basic imaging concepts
02-mri-intro	MRI fundamentals
03-imc-intro	Imaging Mass Cytometry

Connection to Individual Project

Computational imaging techniques can be applied to your individual project – consider MRI analysis, segmentation, or IMC data.